```
import os
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import seaborn as sns

# 1. Raw Dataset Create Karna (Jisme Missing Values aur Duplicates Hon)
np.random.seed(42)
n_rows = 150

data = {
    "Employee_ID": np.random.randint(1001, 1020, size=n_rows),  # Duplicates generate
karne ke liye
    "Department": np.random.choice(["IT", "Sales", "HR", "Marketing", None],
size=n_rows, p=[0.3, 0.3, 0.1, 0.2, 0.1]),
    "Salary_USD": np.random.choice([50000, 65000, 80000, 120000, np.nan],
size=n_rows, p=[0.2, 0.3, 0.3, 0.1, 0.1]),
    "Performance_Score": np.random.randint(1, 11, size=n_rows)
}

df_raw = pd.DataFrame(data)
print("--- Raw Data Ke Shuruati Rows ---")
print(df_raw.head(10))

# 2. Data Cleaning Process
print("
--- Data Cleaning Shuru Ho Rahi Hai ---")

# Step A: Missing Values Handle Karna
print(f"Cleaning se pehle missing values:
{df_raw.isnull().sum()}")
# Numeric columns ko median se aur categorical ko mode se fill karna
df_raw["Salary_USD"] = df_raw["Salary_USD"].fillna(df_raw["Salary_USD"].median())
df_raw["Department"] = df_raw["Department"].fillna(df_raw["Department"].mode()[0])

# Step B: Duplicate Rows Remove Karna
print(f"Total rows cleaning se pehle: {len(df_raw)}")
df_cleaned = df_raw.drop_duplicates(subset=["Employee_ID"],
keep="first").reset_index(drop=True)
print(f"Total rows duplicates hatane ke baad: {len(df_cleaned)}")

# Step C: Inconsistent Data Check (Fixing types if needed)
df_cleaned["Salary_USD"] = df_cleaned["Salary_USD"].astype(int)

# 3. Automated Reporting & Summarization
print("
--- Automated Report Summary ---")
summary_report = df_cleaned.groupby("Department").agg(
    Total_Employees=("Employee_ID", "count"),
    Average_Salary=("Salary_USD", "mean"),
    Avg_Performance=("Performance_Score", "mean")
).reset_index()

print(summary_report)
```

```python
# Summary ko CSV text report ke roop me auto-save karna
summary_report.to_csv("automated_department_report.csv", index=False)

# 4. Reporting Visualization (Automated Summary Plot)
plt.figure(figsize=(10, 5))
sns.barplot(
    x="Department",
    y="Average_Salary",
    data=summary_report,
    palette="Blues_d"
)

plt.title("Automated Summary Report - Average Salary by Department", fontsize=14)
plt.xlabel("Department", fontsize=12)
plt.ylabel("Average Salary ($)", fontsize=12)
plt.grid(True, linestyle="--", alpha=0.5)

# Visual Summary Plot ko save karna submission ke liye
plt.savefig("automated_report_summary.png")
print("
[Success] Report 'automated_department_report.csv' aur visualization
'automated_report_summary.png' save ho chuke hain!")
plt.show()
```